MINISTRY OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH

KURDISTAN REGIONAL GOVERNMENT

Design, Planning, and Installation of Hybrid and On-Grid Solar Photovoltaic Systems for Residential Applications

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A Graduation Research Project
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for the Degree of Bachelor of Science

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Chapter 1

INTRODUCTION

1.1 Background

The escalating global demand for electricity, driven by rapid population growth, urbanization, and industrial expansion, has placed unprecedented pressure on conventional power generation infrastructure worldwide. According to the International Energy Agency [17], global electricity consumption surpassed 29,000 TWh in 2024, with projections indicating a continued annual growth rate of 3.4% through 2030. This rising demand, coupled with the urgent need to mitigate climate change and reduce greenhouse gas (GHG) emissions, has catalyzed a paradigm shift toward renewable energy sources, with solar photovoltaic (PV) technology emerging as one of the most scalable, cost-effective, and rapidly deployable solutions available [18].

The global solar PV market has witnessed extraordinary expansion in recent years. Cumulative installed PV capacity exceeded 2.2 terawatts (TW) by the end of 2024 and reached approximately 2.8 TW by mid-2025, with annual capacity additions surpassing 600 GW—a figure that would have been inconceivable a decade earlier [14, 38]. Solar PV alone accounted for approximately 7% of global electricity generation in 2024, increasing to 8.8% in the first half of 2025, and for the first time in history, a modern renewable energy source led absolute global energy demand expansion [4, 35]. This remarkable trajectory has been fueled by an unprecedented decline in technology costs: the price per watt of PV modules has fallen by approximately 85–90% since 2010, while the installed cost of residential systems has decreased from \$8.00–\$9.00/W to approximately \$2.50–\$3.50/W over the same period [30, 32]. The global benchmark levelized cost of electricity (LCOE) for utility-scale solar reached \$0.039/kWh in 2025 [2], making solar energy cheaper than fossil-fuel-generated electricity in the majority of global markets.

Within this global context, distributed residential PV systems have emerged as a particularly significant segment. Distributed solar installations added nearly 200–220 GW globally in

2024 alone [38]. Residential PV systems enable electricity generation at the point of consumption, reducing transmission losses, enhancing energy security, and empowering households with greater control over their energy costs and carbon footprint. The proliferation of residential solar has been accelerated by supportive policy mechanisms such as net metering, feed-in tariffs, tax incentives, and declining battery storage costs [33, 36].

The Republic of Iraq, with a population of approximately 46.1 million as of 2025 [39], faces one of the most severe and persistent electricity crises in the Middle East and North Africa (MENA) region. Iraq's energy sector is almost entirely dependent on fossil fuels, with oil and natural gas accounting for approximately 99% of electricity generation [5]. The country's CO_2 emissions reached approximately 233 million tonnes in 2023, rising to 239 million tonnes in 2024, with a per capita emission rate of approximately 5.19 tonnes [46]. Despite government plans to increase power generation capacity to 44 GW, Iraq continues to face a persistent supply–demand gap estimated at approximately 10,000 MW, with technical and commercial losses accounting for 50–60% of generated electricity [40]. The residential sector is the largest consumer of electricity in Iraq, accounting for approximately 65% of total national consumption [17], driven primarily by space cooling demands during extreme summer temperatures exceeding 43°C.

The Kurdistan Region of Iraq (KRI), and its capital Erbil in particular, presents a unique case study for residential solar energy deployment. Located at approximately 36.19°N latitude and 44.01°E longitude, at an elevation of approximately 420 m above sea level, Erbil experiences a hot semi-arid climate. The city receives an annual average of approximately 9 hours of sunshine per day, with summer sunshine exceeding 12–14 hours daily. The annual global horizontal irradiation (GHI) in the Kurdistan Region ranges between 1,800 and 2,390 kWh/m^2/year, with daily GHI values of 4.4–5.6 kWh/m^2/day [9], representing exceptionally favorable conditions for photovoltaic energy harvesting. The Kurdistan Regional Government (KRG) has set an ambitious target of achieving 3 GW of solar power capacity by 2033 and launched the "Runaki" (Light) project in late 2024 [22].

Historically, Iraqi households have been heavily reliant on private diesel generators. Households subscribe to neighborhood-based generator services at costs of 10,000–30,000 Iraqi Dinars per ampere per month, with total monthly expenditures reaching 40,000–200,000 IQD ($30–$150 USD) depending on seasonal demand [16, 40]. These diesel generators constitute a significant source of air pollution (CO_2, CO, NO_x, SO_x, and particulate matter), noise pollution, and soil contamination. The transition from diesel generator dependency to residential solar PV systems therefore represents both an economic imperative and an environmental and public health necessity.

Two primary configurations of residential solar PV systems are commonly deployed: on-grid (grid-tied) systems and hybrid systems. On-grid systems operate in direct connection with the utility network, allowing households to consume generated solar electricity while rely-

ing on the grid during periods of insufficient production [25]. These systems are characterized by lower capital costs but automatically shut down during grid outages [6]. In contrast, hybrid PV systems integrate battery energy storage systems (BESS) with grid connectivity, enabling households to store excess solar generation, implement peak-shaving strategies, and maintain power supply during grid interruptions [47]. While hybrid systems incur higher capital costs, they offer substantially greater energy autonomy and supply reliability [26].

Given the unique intersection of abundant solar resources, chronic electricity deficits, heavy reliance on polluting diesel generators, and an evolving regulatory landscape in the Kurdistan Region, this research undertakes a comprehensive comparative analysis of hybrid and on-grid solar PV systems for residential applications in Erbil.

1.2 Literature Review

1.2.1 Evolution of Solar PV Technology and Cost Trends

The photovoltaic effect was first observed by Alexandre-Edmond Becquerel in 1839. Modern PV cells are predominantly manufactured from crystalline silicon [11]. Commercially available monocrystalline silicon cells achieve efficiencies of 18–24%, while polycrystalline cells typically achieve 16–20% [30, 33].

The transition from PERC technology to next-generation architectures—TOPCon, HJT, and IBC—has pushed commercial efficiencies to new heights [24]. Perovskite-on-silicon tandem solar cells have achieved certified laboratory efficiencies exceeding 34%, surpassing the Shockley–Queisser limit for single-junction silicon cells [24, 34]. Modern PV modules exhibit median degradation rates of approximately 0.5–0.6% per year, ensuring systems retain approximately 87–90% of original output after 25 years [19, 31].

1.2.2 Residential On-Grid PV Systems

On-grid systems have been widely studied for residential applications due to their simplicity and lower upfront costs. Rawat et al. [37] demonstrated that on-grid systems effectively meet household demand at significantly lower installation costs than hybrid alternatives. Hassan [12] found that on-grid systems offer lower life-cycle costs in urban settings with reliable grid availability.

However, anti-islanding protection causes on-grid systems to disconnect during outages. This limitation has been identified as a critical barrier in developing countries with unreliable electricity infrastructure [45, 48].

1.2.3 Hybrid PV Systems with Battery Energy Storage

Hybrid PV-battery systems enable excess solar generation to be stored and dispatched during evening peak demand or grid outages [47]. Shah et al. [43] found that hybrid configurations significantly reduce energy bills and carbon footprints. Żelazna et al. [48] concluded that hybrid PV systems are highly competitive when environmental and technological factors are weighted alongside economic considerations. Serat et al. [41] identified polycrystalline hybrid PV systems paired with battery storage as the most cost-effective configuration for energy autonomy.

Battery sizing and optimization are critical determinants. Hesse et al. [13] demonstrated that optimal battery sizing must account for specific household load profiles. Li et al. [23] found that optimization can reduce grid imports by 60–80%. Koskela et al. [21] showed that proper battery sizing can increase self-consumption by over 30%. Gabr et al. [8] demonstrated that multi-objective optimization results in significantly shorter payback periods. Khezri et al. [20] provided a universal methodology using Net Present Cost minimization. Shabbir et al. [42] confirmed that detailed load-curve analysis is necessary for optimal battery sizing.

1.2.4 Solar Energy Potential and Electricity Context in Iraq

Despite receiving annual GHI values of 1,800–2,390 kWh/m^2/year and approximately 2,979 hours of annual sunshine, renewable energy accounted for only 1–2% of Iraq's total electricity generation as of 2025 [5, 18]. The federal government has set a target of 12 GW of renewable energy by 2030. However, existing Iraqi solar research has focused on theoretical assessment rather than practical field performance evaluation.

1.2.5 System Design Methodologies

Residential PV system design involves load assessment, system sizing, component selection, and installation optimization [1]. System sizing must account for daily solar irradiation, module efficiency, temperature derating, system losses, and desired energy autonomy [29]. For hybrid systems, battery capacity determination requires consideration of daily energy consumption, backup duration, depth of discharge limits, and cycle life [42]. The ideal tilt angle for Erbil approximates 30–35°.

1.3 Research Gap

Despite the substantial body of literature surrounding photovoltaic systems, several critical gaps persist within the specific context of the Kurdistan Region of Iraq:

1. **Lack of field-collected data:** The majority of existing regional studies rely heavily on computational simulations and theoretical modeling, with a marked deficit in research utilizing actual measurements under Erbil's specific climatological conditions.

2. **Generic load profiles:** Most published analyses employ idealized residential load profiles that fail to capture the unique, highly volatile consumption patterns of Iraqi households, characterized by extreme summer cooling demands and diesel generator rotations.

3. **Absence of direct comparisons:** There is a scarcity of literature presenting direct, side-by-side empirical comparisons of on-grid and hybrid configurations operating under identical environmental conditions in Iraq.

4. **Unquantified diesel displacement:** Existing literature rarely quantifies the specific economic and environmental benefits of residential PV in terms of displacing the highly expensive, unregulated private diesel generator sector.

5. **Evolving regulatory context:** The rapidly changing regulatory landscape in the Kurdistan Region, including emerging policies like the "Runaki" initiative, has not yet been adequately incorporated into localized PV system design frameworks.

1.4 Objectives of the Study

The primary aim is to develop a practical, locally grounded framework for the design, performance evaluation, and techno-economic comparison of hybrid and on-grid residential PV systems in Erbil. Specific objectives:

1. To collect and analyze field-measured experimental data on PV system parameters under real operating conditions in Erbil.

2. To design and size residential on-grid and hybrid PV systems based on measured performance data and actual household energy demand.

3. To evaluate and compare on-grid and hybrid system performance in terms of energy yield, efficiency, capacity factor, and reliability.

4. To conduct a comprehensive techno-economic analysis comparing capital costs, operational costs, payback periods, and LCOE for both configurations.

5. To provide evidence-based, implementable recommendations for optimal residential PV system deployment in the Kurdistan Region.

1.5 Novelty and Contribution

This research introduces several novel contributions to the field of renewable energy systems engineering, specifically tailored to the Middle Eastern operational environment:

- **Field-based experimental validation:** By utilizing empirical data collected from an active, real-world PV installation in Erbil, the study transcends theoretical modeling to provide grounded, real-world performance characterizations.
- **Dual-configuration comparative framework:** The research enables the simultaneous, rigorous evaluation of both hybrid and on-grid architectures under identical environmental and localized load conditions.
- **Context-specific economic analysis:** The study uniquely integrates Erbil's complex multi-tiered household consumption patterns, utility tariff structures, and the granular economics of diesel generator displacement.
- **Contribution to a data-scarce region:** This work contributes vital primary experimental data to a geographically underserved research domain, establishing a new baseline for future studies in the region.

1.6 Significance of the Study

The findings of this study carry significant practical and academic implications across multiple stakeholder levels:

- **For residential consumers and homeowners:** The techno-economic frameworks provide actionable, evidence-based guidance for selecting the optimal PV system configuration, facilitating informed capital investments.
- **For energy policymakers and the KRG:** The empirical data and grid-interaction analyses offer critical insights needed to design more effective incentive programs and net-metering regulations, supporting the 3 GW by 2033 mandate.
- **For system designers and EPC contractors:** The study provides a comprehensive reference of locally validated design parameters, sizing methodologies, and performance benchmarks that mitigate the risk of system underperformance.
- **For the academic community:** The research enriches the broader renewable energy discourse by injecting high-quality, primary field data from an underrepresented, high-temperature semi-arid climate zone.

Chapter 2

SYSTEM COMPONENTS AND WORKING PRINCIPLES

2.1 Overview

The design and implementation of residential photovoltaic (PV) systems—whether on-grid or hybrid—require an in-depth understanding of the individual components, their operating principles, and how they interact. This chapter presents a detailed examination of the core components: solar PV modules, inverters, battery energy storage systems, and grid connection mechanisms.

2.2 Solar Photovoltaic (PV) Modules

2.2.1 Operating Principle: The Photovoltaic Effect

Solar PV modules convert incident solar radiation directly into electrical energy through the photovoltaic effect—a quantum mechanical process first observed by Becquerel in 1839 and first practically demonstrated by Bell Laboratories in 1954 [3].

The photovoltaic effect occurs within semiconductor materials—most commonly crystalline silicon (Si)—engineered to contain a p-n junction. When photons strike the semiconductor surface with energy greater than the material's bandgap energy (approximately 1.12 eV for silicon), they excite valence electrons into the conduction band, creating electron-hole pairs. The internal electric field at the p-n junction separates these charge carriers: electrons are swept toward the n-type (phosphorus-doped) layer, while holes migrate toward the p-type (boron-doped) layer. This charge separation establishes a potential difference across the cell, and when an external circuit is connected, current flows [10, 27].

figures/fig2_1_pv_conversion.png

Figure 2.1: The conversion of solar radiation to electricity in PV systems, showing the p-n junction, photon absorption, electron flow, and hole flow mechanisms.

The performance of a PV cell is characterized by several key electrical parameters measured under Standard Test Conditions (STC: 1000 W/m^2 irradiance, 25°C cell temperature, and an Air Mass of 1.5):

- **Open-Circuit Voltage (V_{oc}):** This represents the maximum voltage available from a solar cell when no current is flowing, typically ranging from 0.5 to 0.7 V for standard silicon cells.
- **Short-Circuit Current (I_{sc}):** This is the maximum current output at zero voltage, which is directly proportional to the incident solar irradiance and the active cell area.
- **Maximum Power Point (P_{max}):** During normal operation, the cell operates at this specific voltage and current point where the electrical power output is maximized.
- **Fill Factor (FF):** Defined as the ratio of P_{max} to the product of V_{oc} and I_{sc}. High-quality cells typically exhibit a fill factor between 0.75 and 0.85.
- **Conversion Efficiency (η):** This dictates the proportion of incident solar power that is successfully converted into usable electrical power.

The fundamental relationships are:

$$P_{max} = V_{mp} \times I_{mp} = FF \times V_{oc} \times I_{sc} \tag{2.1}$$

$$\eta = \frac{P_{max}}{G \times A} \times 100\% \tag{2.2}$$

where G is solar irradiance (W/m^2) and A is cell area (m^2).

figures/fig2_2_iv_curve.png

Figure 2.2: Typical I-V and P-V characteristic curves of a solar cell showing open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), and the maximum power point (P_{max}).

2.2.2 PV Module Construction and Types

A PV module consists of multiple cells electrically connected in series and parallel to achieve desired voltage and current ratings. Cells are encapsulated between a front glass cover and rear backsheet with EVA encapsulant, providing mechanical protection and moisture resistance.

Table 2.1: Comparison of Major PV Module Technologies

Parameter	**Monocrystalline Si**	**Polycrystalline Si**	**Thin-Film**
Cell Efficiency	18–24%	16–20%	10–13%
Module Efficiency	17–22%	15–18%	9–12%
Temp. Coefficient	–0.3 to –0.4 %/°C	–0.4 to –0.5 %/°C	–0.2 to –0.3 %/°C
Degradation Rate	0.3–0.5 %/yr	0.5–0.7 %/yr	0.5–1.0 %/yr
Lifespan	25–30+ years	25–30 years	20–25 years
Space Efficiency	High	Medium	Low
Cost per Watt	Moderate–High	Low–Moderate	Low

Monocrystalline silicon modules dominate the residential market due to superior efficiency and favorable temperature coefficients. The transition from PERC to TOPCon and HJT architectures has further improved efficiencies while reducing degradation [11, 24].

2.2.3 Environmental Factors Affecting Performance

The actual field output of a PV module is highly dependent on localized environmental parameters that cause deviations from STC ratings:

- **Operating Temperature:** Module efficiency exhibits a negative correlation with increasing cell temperature, typically degrading at a rate of –0.3 to –0.5 %/°C. In Erbil's extreme summer climate, where rooftop module temperatures can reach 60–70°C, temperature-induced power losses can reach 15–20% relative to baseline STC values [44].
- **Soiling and Dust Accumulation:** The accumulation of airborne particulate matter poses a significant challenge. Studies in arid Middle Eastern climates indicate soiling can induce yield losses of 15–30% if regular maintenance is neglected, making periodic cleaning essential during the Kurdistan Region's dry summers [28].
- **Wind Speed:** While moderate wind velocities can marginally improve performance by providing convective cooling to the modules, extreme wind loads necessitate robust structural mounting to prevent mechanical failure.
- **Shading:** Even partial shading from nearby structures or vegetation can disproportionately reduce total array output. Because cells within a module are series-connected, a shaded cell acts as a reverse-biased resistor, restricting current flow and creating localized hot spots.

figures/fig2_3_rooftop_orientation.png

Figure 2.3: Typical residential rooftop solar PV installation showing panel arrangement, mounting structure, tilt angle, and exposure to sunlight with minimal shading conditions.

2.3 Inverters

2.3.1 Function and Operating Principle

The primary function of a solar inverter is the conversion of direct current (DC) electricity generated by the PV array into alternating current (AC). However, modern inverters act as the central intelligence hub of the PV system, executing several critical ancillary functions:

- **Maximum Power Point Tracking (MPPT):** The inverter continuously samples the voltage and current of the PV array and adjusts the electrical operating point to ensure maximum power extraction under fluctuating conditions, frequently achieving tracking efficiencies exceeding 99% [7].
- **Grid Synchronization:** For grid-tied operations, the inverter precisely matches the output voltage, phase, and frequency to the local utility specifications (50 Hz in Iraq) while maintaining Total Harmonic Distortion (THD) below 3–5%.

- **Anti-Islanding Protection:** Crucially for safety, grid-interactive inverters must rapidly detect utility outages and automatically disconnect from the grid per IEEE 1547 standards [15], preventing power back-feed into compromised distribution lines.
- **System Monitoring:** Contemporary inverters integrate robust data logging and communication protocols (Wi-Fi, Ethernet, RS485), enabling real-time system monitoring, fault diagnostics, and remote management.

2.3.2 Types of Inverters

Table 2.2: Comparison of Inverter Types for Residential PV Systems

Parameter	String	Microinverter	Hybrid
Peak Efficiency	96–98%	95–97%	93–97%
MPPT Level	System (1–2)	Module-level	System + Battery
Shading Tolerance	Low	High	Low–Medium
Battery Integration	No	No	Yes (built-in)
Islanding Capability	No	No	Yes
Cost per Watt	Lowest	Highest	Moderate–High
Typical Lifespan	10–15 yr	15–25 yr	10–15 yr

For this study, a hybrid inverter is employed for the hybrid system, while a standard string inverter is utilized for the on-grid system.

2.4 Battery Energy Storage Systems (BESS)

2.4.1 Role of Battery Storage

Within a hybrid PV architecture, Battery Energy Storage Systems (BESS) are utilized to resolve the fundamental temporal mismatch between peak solar generation and residential consumption, serving three primary strategic functions:

- **Energy Time-Shifting:** By storing excess diurnal generation for nocturnal use, households can increase their volumetric self-consumption of solar energy from a baseline of 30–40% to upwards of 60–80% [21].
- **Backup Power:** In the context of the Kurdistan Region's unreliable electricity infrastructure, batteries provide uninterruptible power supply during sudden utility grid outages.

- **Peak Shaving:** Intelligent hybrid inverters can utilize stored battery capacity to deliberately discharge during periods of high grid demand, minimizing reliance on expensive utility tariffs or neighborhood diesel generator subscriptions.

figures/fig2_4_wiring_diagram.png

Figure 2.4: Electrical wiring and connections in a residential PV system showing integration between solar panels, charge controller, battery, inverter, and load.

2.4.2 Battery Technologies

Table 2.3: Comparison of Battery Technologies for Residential PV Systems

Parameter	**Lead-Acid**	**Li-Ion (NMC)**	**LiFePO4 (LFP)**
Energy Density	30–50 Wh/kg	150–220 Wh/kg	90–160 Wh/kg
Cycle Life (80% DoD)	300–800	2,000–4,000	3,000–10,000
Recommended DoD	50%	80–90%	80–100%
Round-Trip Efficiency	75–85%	90–95%	92–98%
Calendar Life	3–5 yr	8–12 yr	10–15 yr
Safety	H_2 off-gassing	Thermal runaway risk	Excellent stability
Cost per kWh (2024)	$100–200	$250–400	$200–350

LFP chemistry is increasingly preferred for residential applications due to excellent thermal stability, non-toxicity, and extended cycle life [13, 47].

2.4.3 Battery Sizing

The required battery capacity can be estimated as:

$$C_{batt} = \frac{E_{night} \times D_{autonomy}}{DoD \times \eta_{batt}} \tag{2.3}$$

where:

- E_{night} = daily energy consumption during non-solar hours (Wh)
- $D_{autonomy}$ = desired days of autonomous operation (typically 1–2)
- DoD = maximum depth of discharge (0.8 for LFP, 0.5 for lead-acid)
- η_{batt} = battery round-trip efficiency (0.90–0.95 for Li-ion)

Research by Khezri et al. [20] and Hesse et al. [13] demonstrated that optimal sizing requires detailed household load profile analysis rather than generic rule-of-thumb approaches.

2.5 Grid Connection and Net Metering

2.5.1 Grid Interconnection

Grid interconnection enables bidirectional power flow between the PV system and the utility. Compliance with IEEE 1547 [15] and IEC 61727 is required, specifying voltage/frequency operating ranges, harmonic limits, anti-islanding response, and grounding requirements.

2.5.2 Net Metering

Net metering credits PV system owners for surplus exported electricity using a bidirectional meter. In the Iraqi context, net metering policies are still in early development. The absence of mature net metering regulations currently limits purely on-grid system economics—further strengthening the case for hybrid systems with self-consumption optimization.

2.6 System Architectures: On-Grid vs. Hybrid

2.6.1 On-Grid System Architecture

An on-grid system consists of the PV array, grid-tied inverter, and balance-of-system components:

figures/fig2_5_safety_components.png

Figure 2.5: Safety components in a residential PV system including grounding, protection devices, DC disconnect, inverter, and emergency disconnect mechanisms.

Key characteristics: no battery storage, system ceases during outages, lowest capital cost, highest conversion efficiency (no battery losses).

2.6.2 Hybrid System Architecture

Hybrid systems add battery storage and a hybrid inverter. Unlike standard grid-tied configurations, the hybrid inverter is programmed with a dynamic energy management algorithm that prioritizes power flows to maximize self-reliance and economic savings:

1. **Supply immediate loads:** Available solar generation is first directed to satisfy immediate household electrical loads.
2. **Charge the battery bank:** Any excess generation is subsequently routed to charge the batteries.

3. **Export surplus power:** Only once the batteries reach a fully charged state is surplus power exported to the utility grid.

4. **Discharge batteries:** During periods of insufficient solar irradiance or at night, the system defaults to discharging the batteries to meet household demand.

5. **Import from utility:** The grid acts strictly as a tertiary source of last resort, supplying power only when both solar generation and battery reserves are depleted.

In the event of a utility grid failure, an internal transfer switch rapidly isolates the system from the grid and transitions the inverter into island mode, allowing it to continue supplying critical household loads from the PV array and battery reserves—a decisive operational advantage in regions plagued by chronic infrastructure instability.

Table 2.4: Comparative Summary: On-Grid vs. Hybrid System Architectures

Feature	**On-Grid**	**Hybrid**
Battery Storage	None	Yes (Li-ion or lead-acid)
Grid Dependency	High	Medium
Self-Consumption	30–40%	60–80%
Capital Cost	Lower	Higher (30–60% premium)
Complexity	Low	Medium–High
Energy Independence	Low	Medium–High
During Outages	Shuts down	Operates (island mode)
Conversion Efficiency	Higher	Slightly lower (5–10% battery losses)
Suitability for Erbil	Limited	Highly suitable

2.7 Chapter Summary

This chapter examined the core components of residential PV systems: (1) PV modules and the photovoltaic effect, (2) inverter types and MPPT technology, (3) battery energy storage technologies and sizing, (4) grid interconnection and net metering, and (5) on-grid versus hybrid architectures. These foundations support the design calculations and performance analysis in subsequent chapters.

Chapter 3

METHODOLOGY: SYSTEM DESIGN, PLANNING, AND DATA COLLECTION

3.1 Overview

This chapter presents the complete methodology employed in this study, encompassing the site characterization, residential load assessment, theoretical system design and sizing calculations for both on-grid and hybrid PV configurations, equipment specifications, installation procedures, and the experimental data collection protocol. All design calculations are grounded in Erbil's specific climatic conditions, Iraqi residential building characteristics, and locally measured performance data.

3.2 Study Site Characterization

3.2.1 Geographic and Climatic Parameters

The experimental PV system was installed on a residential building located in Erbil, the capital of the Kurdistan Region of Iraq. Table 3.1 summarizes the key geographic and climatic parameters of the study site.

Table 3.1: Geographic and Climatic Parameters of the Study Site

Parameter	**Value**
Location	Erbil, Kurdistan Region, Iraq
Latitude	36.19°N
Longitude	44.01°E
Elevation	∼420 m above sea level
Climate Classification	Hot semi-arid (Köppen: BSh)
Annual Average Temperature	20–22°C
Summer Peak Temperature	43–50°C (July–August)
Winter Minimum Temperature	0–5°C (January)
Annual Sunshine Hours	∼2,979 hours
Daily Sunshine (Summer)	12–14 hours
Annual GHI	1,800–2,390 kWh/m^2/year
Daily Average GHI	4.4–5.6 kWh/m^2/day
Peak Sun Hours (PSH)	5.0–5.5 hours/day (annual avg.)
Prevailing Wind	NW, 2–4 m/s average
Dust/Soiling Season	May–October (dry, minimal rainfall)

figures/fig3_1_erbil_solar_map.png

Figure 3.1: Solar irradiance map of the Kurdistan Region showing GHI distribution. Source: Global Solar Atlas / World Bank ESMAP [9].

3.2.2 Iraqi Residential Building Characteristics

Residential buildings in Erbil are predominantly constructed from reinforced concrete frames with hollow concrete block infill walls. The typical single-family dwelling occupies a plot area of 200–300 m^2, with a usable flat rooftop area of approximately 100–200 m^2 after accounting for stairwell enclosures, water tanks, and satellite dishes. The flat concrete rooftop construction prevalent in Iraqi architecture is highly suitable for solar PV installation, as it:

- Provides a structurally robust, load-bearing surface capable of supporting PV module and mounting system weight (typically 15–25 kg/m^2).
- Eliminates the need for specialized roof-penetrating attachments required on pitched roofs.
- Allows flexible module orientation and tilt angle adjustment using ground-mounted racking systems.

- Offers unobstructed southern exposure in most residential neighborhoods, as buildings are typically 2–3 stories with setbacks.

For this study, the available rooftop area was measured at approximately 120 m^2, with an effective usable area of approximately 80 m^2 after deducting obstructions.

3.3 Residential Load Assessment

3.3.1 Household Electrical Load Profile

An accurate assessment of household electrical demand is the foundational step in PV system sizing. The study residence represents a typical middle-class Erbil household. Table 3.2 presents the detailed load inventory.

Table 3.2: Detailed Household Electrical Load Inventory

Appliance	**Qty**	**Rated (W)**	**Daily Hours**	**Daily (Wh)**
LED Lighting	12	10	6	720
Ceiling Fans	4	75	8	2,400
Refrigerator	1	150	24 (cycling)	1,800
Television (LED)	2	80	5	800
Washing Machine	1	500	1	500
Water Pump	1	750	2	1,500
Internet Router	1	15	24	360
Phone Chargers	4	10	3	120
Laptop	2	65	4	520
Air Cooler (Evap.)	2	200	10	4,000
Split AC (Summer)	1	1,500	8	12,000
Iron	1	1,200	0.5	600
Miscellaneous	–	200	3	600
Winter Total				**∼9,320 Wh**
Summer Total				**∼25,920 Wh**

The seasonal disparity is significant: summer consumption is approximately 2.8× higher than winter due to space cooling loads. For conservative system design, the annual average daily consumption is estimated at approximately 15,000 Wh/day (15 kWh/day), accounting for seasonal weighting.

3.3.2 Grid Supply and Diesel Generator Context

A critical contextual factor in Erbil is that the utility grid does not provide 24-hour continuous supply. Typical grid availability ranges from 12–18 hours per day, with scheduled and unscheduled outages occurring daily. During outages, households rely on neighborhood-based private diesel generators, subscribing at costs of 10,000–30,000 IQD per ampere per month [40]. A typical household subscribing to 10–15 amperes incurs monthly generator costs of approximately 100,000–450,000 IQD ($75–$340 USD).

This dual-supply reality fundamentally shapes the design rationale: a hybrid PV system with battery storage can directly displace diesel generator dependency, while an on-grid system cannot provide power during grid outages when generator backup is needed most.

3.4 PV System Sizing and Design Calculations

3.4.1 Optimal Tilt Angle and Orientation

For fixed-tilt installations in the Northern Hemisphere, the optimal annual tilt angle (β_{opt}) is commonly approximated as:

$$\beta_{opt} \approx \phi \pm (10^\circ \text{ to } 15^\circ) \tag{3.1}$$

where ϕ is the site latitude. For Erbil ($\phi = 36.19°$N):

- **Annual optimum:** $\beta_{opt} \approx 30°$–$35°$ (south-facing)
- **Summer optimum:** $\beta_{opt} \approx 15°$–$20°$ (lower tilt to capture high-altitude sun)
- **Winter optimum:** $\beta_{opt} \approx 45°$–$50°$ (steeper tilt for low-altitude sun)

For this study, a fixed tilt angle of $\beta = 33°$ facing due south (azimuth = 180°) was selected as the year-round compromise, consistent with Global Solar Atlas recommendations for this latitude [9].

figures/fig3_2_tilt_orientation.png

Figure 3.2: Schematic illustration of PV module tilt angle (β) and azimuth orientation for the study site in Erbil, with the optimal south-facing configuration at 33° tilt.

3.4.2 PV Array Sizing

The required PV array capacity is determined by the daily energy demand, available solar resource, and system losses:

$$P_{PV} = \frac{E_{daily}}{PSH \times \eta_{sys}} \tag{3.2}$$

where:

- E_{daily} = average daily energy consumption = 15,000 Wh = 15 kWh
- PSH = Peak Sun Hours = 5.0 h/day (annual average for Erbil)
- η_{sys} = overall system efficiency factor

The system efficiency factor accounts for cumulative losses:

Table 3.3: System Loss Budget for PV Array Sizing

Loss Factor	**Value**	**Efficiency**
Temperature derating	10–15%	0.875
Soiling/dust	5–10%	0.925
Module mismatch	2%	0.98
DC wiring losses	2%	0.98
Inverter efficiency	3–5%	0.96
AC wiring losses	1%	0.99
Combined η_{sys}		$\approx$ **0.75**

Substituting into Equation 3.2:

$$P_{PV} = \frac{15{,}000}{5.0 \times 0.75} = \frac{15{,}000}{3.75} = 4{,}000 \text{ W} = 4.0 \text{ kWp} \tag{3.3}$$

A minimum PV array capacity of **4.0 kWp** is required. To provide a safety margin and account for degradation over the system lifetime, a design capacity of **5.0 kWp** is recommended.

3.4.3 Module Selection and Array Configuration

Based on the 5.0 kWp design target, the following module configuration is proposed:

Table 3.4: PV Module Specifications and Array Configuration

Parameter	**Specification**
Module Technology	Monocrystalline Si (TOPCon)
Module Rating	550 Wp (STC)
Module Efficiency	∼21.3%
V_{oc} per module	∼49.6 V
I_{sc} per module	∼14.0 A
V_{mp} per module	∼41.7 V
I_{mp} per module	∼13.2 A
Module Dimensions	∼2278 × 1134 × 30 mm
Weight per module	∼28.5 kg
Temp. Coefficient (P_{max})	–0.34%/°C
Number of modules	9 modules (9 × 550 = 4,950 Wp)
Array configuration	1 string × 9 modules in series
String V_{oc}	9 × 49.6 = 446.4 V
String I_{sc}	14.0 A
Total array area	9 × 2.58 = 23.3 m^2

Note: The above specifications represent the design-stage component selection. The measured experimental data in Table 3.6 reflects the actual field performance of the installed system under varying environmental conditions.

3.4.4 Battery Sizing (Hybrid Configuration Only)

For the hybrid system, battery capacity is sized to provide backup power during evening/nighttime hours and grid outages:

$$C_{batt} = \frac{E_{backup} \times D_{autonomy}}{DoD \times \eta_{batt} \times V_{sys}} \tag{3.4}$$

where:

- E_{backup} = energy required during non-solar hours ≈ 8,000 Wh (estimated 53% of daily consumption occurs between 6 PM and 8 AM)
- $D_{autonomy}$ = 1 day (minimum backup duration)

- DoD = 0.80 (for $LiFePO_4$ batteries)
- η_{batt} = 0.95 (round-trip efficiency)
- V_{sys} = 48 V (system bus voltage)

$$C_{batt} = \frac{8{,}000 \times 1}{0.80 \times 0.95 \times 48} = \frac{8{,}000}{36.48} \approx 219 \text{ Ah} \tag{3.5}$$

A battery bank of **200–250 Ah at 48 V** (approximately 10–12 kWh usable capacity) is recommended. This can be achieved using a single 48V 200Ah $LiFePO_4$ rack-mounted battery or equivalent configuration.

3.4.5 Inverter Selection

Table 3.5: Inverter Specifications for On-Grid and Hybrid Configurations

Parameter	**On-Grid Inverter**	**Hybrid Inverter**
Type	Grid-tied string	Hybrid (battery + grid)
Rated AC Output	5 kW	5 kW
Max. DC Input	6.5 kWp	6.5 kWp
MPPT Channels	2	2
MPPT Voltage Range	150–500 V DC	150–500 V DC
Max. Input Voltage	600 V DC	600 V DC
Battery Voltage	–	48 V
Max. Charge Current	–	100 A
Peak Efficiency	97.5%	96.5%
Anti-Islanding	Yes (IEEE 1547)	Yes + Island Mode
Communication	Wi-Fi / RS485	Wi-Fi / RS485
Grid Frequency	50 Hz	50 Hz

3.5 System Architecture and Wiring

3.5.1 On-Grid System Configuration

The on-grid system consists of:

1. PV array (9 × 550 Wp modules in series) connected to a grid-tied string inverter via DC isolator and surge protection.

2. AC output from the inverter connected to the household distribution board through an AC isolator and bidirectional energy meter.
3. Earth grounding system compliant with IEC 62305.

3.5.2 Hybrid System Configuration

The hybrid system includes the same PV array but connects to a hybrid inverter with integrated charge controller:

1. PV array $\rightarrow$ DC combiner box $\rightarrow$ Hybrid inverter (MPPT input).
2. Battery bank (48V $LiFePO_4$) $\rightarrow$ Hybrid inverter (battery port).
3. Hybrid inverter AC output $\rightarrow$ Household distribution board.
4. Grid connection $\rightarrow$ Hybrid inverter (grid port) with bidirectional metering.
5. Automatic Transfer Switch (ATS) for seamless grid/island mode transition.

figures/fig3_3_system_wiring.png

Figure 3.3: Electrical wiring diagram of the hybrid PV system showing connections between PV array, charge controller, battery bank, hybrid inverter, distribution board, and utility grid.

figures/fig3_4_safety_grounding.png

Figure 3.4: Safety and protection components of the PV installation including DC disconnect, surge protection devices (SPD), grounding electrode, and emergency shutdown mechanism.

3.6 Experimental Data Collection

3.6.1 Measurement Protocol

Field measurements were conducted over a multi-day campaign spanning two distinct seasonal periods to capture performance variation:

- **Winter period:** February 12–16, 2026 (5 consecutive days)
- **Spring period:** April 13, 2026 (2 measurement sessions)

For each measurement day, readings were recorded at three time intervals to capture the diurnal performance curve:

- **Morning:** 08:00–09:30 (low solar altitude, warming conditions)
- **Midday:** 12:00–13:30 (peak solar irradiance)

- **Afternoon:** 15:00–16:30 (declining irradiance)

3.6.2 Measured Parameters and Instruments

The following electrical and environmental parameters were recorded at each measurement interval:

- **Ambient Temperature (T_{amb}):** Measured using a calibrated digital thermometer ($\pm$0.5°C accuracy).
- **Wind Speed:** Measured using a handheld anemometer ($\pm$0.5 km/h accuracy).
- **Open-Circuit Voltage (V_{oc}):** Measured at the PV array terminals with load disconnected, using a digital multimeter.
- **Short-Circuit Current (I_{sc}):** Measured by shorting the PV array terminals through a calibrated DC ammeter.
- **Maximum Power Point Voltage (V_{mp}):** Recorded from the inverter MPPT display during connected operation.
- **Maximum Power Point Current (I_{mp}):** Recorded from the inverter MPPT display during connected operation.
- **Maximum Power Output (P_{max}):** Calculated as $P_{max} = V_{mp} \times I_{mp}$, and cross-verified with the inverter power display.

3.6.3 Collected Experimental Data

Table 3.6 presents the complete dataset of field-measured PV system parameters under varying environmental conditions across the experimental campaign.

Table 3.6: Measured PV System Parameters Under Different Environmental Conditions

Date	**Time of Day**	**Temp (°C)**	**Wind (km/h)**	P_{max} **(Wp)**	V_{oc} **(V)**	I_{sc} **(A)**	V_{mp} **(V)**	I_{mp} **(A)**
12-2-2026	Morning	11	12	35.3	46.0	0.8	44.2	0.8
	Midday	15	14	185.2	47.7	4.0	46.3	4.0
	Afternoon	16	16	661.3	46.6	16.7	39.6	16.7
13-2-2026	Morning	10	3.6	569	46.7	12.0	39.7	14.33
	Midday	14	6	601	47.3	12.7	40.0	14.0
	Afternoon	19	6	709.6	45.2	15.7	38.4	15.0
14-2-2026	Morning	11	2	246	47.4	5.2	38.5	4.0
	Midday	14	4	191	45.7	4.2	38.8	5.0
	Afternoon	13	6	199.87	46.7	4.2	39.7	5.0
15-2-2026	Morning	10	7	650	47.4	16.4	41.7	15.0
	Midday	14	10	584	46.2	15.0	40.6	14.4
	Afternoon	10	10	540	45.5	14.0	40.0	13.5
16-2-2026	Morning	12	18	600	46.4	15.4	40.8	14.7
	Midday	14	25	702	47.4	17.5	41.8	16.8
	Afternoon	16	13	560	45.8	14.6	40.3	13.9
13-4-2026	Morning	14	16	643	47.45	17.0	41.7	14.5
(Session 1)	Midday	15	18	832.4	47.15	22.0	41.9	19.83
	Afternoon	6	21	155	45.0	4.3	38.3	4.0
13-4-2026	Morning	13	6	402.2	46.75	10.0	42.3	9.5
(Session 2)	Midday	15	5	622.4	46.2	15.4	42.23	14.7
	Afternoon	15	5	602.3	46.3	16.0	39.59	15.15

3.6.4 Data Quality Observations and Limitations

While the collected dataset provides valuable field-measured performance data under real operating conditions, several anomalous readings were identified during post-processing and are transparently acknowledged as limitations of the experimental campaign:

1. **Anomalous temperature reading (13 April 2026, Session 1, Afternoon):** A recorded ambient temperature of 6°C at approximately 15:00–16:30 in April is climatologically inconsistent with Erbil's typical spring afternoon temperatures, which normally range between 18–28°C during this period. This anomaly is most likely attributable to a sensor error, transient instrument malfunction, or a recording transcription error. The correspondingly low power output (155 Wp) in this row suggests that heavy cloud cover or a

sudden weather event may have coincided with the measurement, but cannot alone explain the temperature deviation. This data point should be treated with caution in subsequent analyses.

2. **Inverted diurnal output profile (14 February 2026):** On this date, the midday power output (P_{max} = 191 Wp) was recorded as lower than the morning output (P_{max} = 246 Wp), which contradicts the expected diurnal irradiance curve where midday solar radiation is at its peak. This pattern is consistent with transient cloud cover or atmospheric particulate interference during the midday measurement window. The uniformly low output across all three intervals (246, 191, and 200 Wp) compared to adjacent days (569–710 Wp) suggests that 14 February was predominantly overcast, representing a worst-case performance scenario.

3. **I_{mp} equal to I_{sc} (12 February 2026, Morning):** The morning measurement on 12 February records both I_{sc} and I_{mp} at 0.8 A. Under standard photovoltaic theory, the maximum power point current (I_{mp}) must always be less than the short-circuit current (I_{sc}), as the operating point shifts along the I-V characteristic curve. The equality of these two values at very low irradiance (corresponding to only 35.3 Wp output) likely reflects measurement resolution limitations of the instruments at near-threshold operating conditions, where the I-V curve becomes nearly flat and the distinction between I_{sc} and I_{mp} falls within the meter's accuracy tolerance.

These anomalies do not invalidate the overall dataset, as the remaining 18 of 21 data points exhibit internally consistent and physically plausible relationships. However, they highlight the inherent challenges of field-based experimental data collection and the importance of instrument calibration, repeated measurements, and multi-day sampling campaigns—practices that were followed in this study to the extent practicable. Future work should incorporate continuous data logging equipment with higher temporal resolution to minimize the impact of such isolated measurement artifacts.

3.7 Performance Evaluation Metrics

The following metrics are employed to evaluate and compare on-grid and hybrid system performance in Chapter 4:

- **Daily Energy Yield (E_{day}):** Total energy generated per day (kWh/day), calculated by integrating power output over sunshine hours.
- **Specific Yield (Y_{sp}):** Energy generated per unit of installed capacity (kWh/kWp/day),

enabling comparison across different system sizes:

$$Y_{sp} = \frac{E_{day}}{P_{rated}} \tag{3.6}$$

- **Performance Ratio (PR):** Ratio of actual energy output to theoretical output under STC, indicating overall system quality:

$$PR = \frac{E_{actual}}{E_{ideal}} = \frac{E_{actual}}{P_{rated} \times PSH} \tag{3.7}$$

- **Capacity Factor (CF):** Ratio of actual energy output to theoretical maximum output if the system operated at rated capacity 24/7:

$$CF = \frac{E_{actual}}{P_{rated} \times 24} \times 100\% \tag{3.8}$$

- **Self-Consumption Ratio (SCR):** Fraction of solar generation consumed directly by the household (relevant for hybrid systems with storage).
- **Self-Sufficiency Ratio (SSR):** Fraction of total household demand met by the PV system (with or without storage).

3.8 Techno-Economic Analysis Framework

The economic comparison between on-grid and hybrid configurations employs the following metrics:

- **Total System Cost:** Capital expenditure (CAPEX) including modules, inverter, batteries (hybrid only), mounting structure, wiring, and installation labor.
- **Levelized Cost of Electricity (LCOE):**

$$LCOE = \frac{C_{total} + \sum_{t=1}^{N} \frac{C_{O\&M,t}}{(1+r)^t}}{\sum_{t=1}^{N} \frac{E_t}{(1+r)^t}} \tag{3.9}$$

  where C_{total} is the initial capital cost, $C_{O\&M,t}$ is the annual operation and maintenance cost, E_t is the annual energy production, r is the discount rate (8%), and N is the system lifetime (25 years).

- **Simple Payback Period (SPP):**

$$SPP = \frac{C_{total}}{S_{annual}} \tag{3.10}$$

where S_{annual} is the total annual savings (avoided grid electricity cost + avoided diesel generator subscription).

- **Diesel Generator Displacement Value:** The monthly cost savings from reducing or eliminating private generator subscriptions, calculated based on local ampere-based pricing.

3.9 Chapter Summary

This chapter presented the complete methodological framework for the study:

1. The study site in Erbil was characterized geographically and climatically, with annual GHI of 1,800–2,390 kWh/m^2/year and Peak Sun Hours of 5.0 h/day.
2. A detailed residential load assessment yielded a design daily consumption of 15 kWh/day, with significant seasonal variation (9.3 kWh winter vs. 25.9 kWh summer).
3. Theoretical PV array sizing using loss-adjusted calculations determined a minimum 4.0 kWp system, with a recommended 5.0 kWp design capacity (9 $\times$ 550 Wp modules).
4. Battery sizing for the hybrid configuration yielded a requirement of 200–250 Ah at 48 V ($\sim$10–12 kWh usable).
5. Field measurements were conducted over 7 days across two seasonal periods (February and April 2026), with 21 data points capturing V_{oc}, I_{sc}, V_{mp}, I_{mp}, P_{max}, temperature, and wind speed.
6. Performance evaluation metrics (energy yield, PR, CF, SCR, SSR) and techno-economic analysis frameworks (LCOE, SPP, diesel displacement) were defined for the comparative analysis in Chapter 4.

Chapter 4

RESULTS, ANALYSIS, AND DISCUSSION

4.1 Overview

This chapter presents the detailed analysis of the field-measured experimental data collected in Chapter 3, followed by the performance evaluation of both on-grid and hybrid PV system configurations, a comprehensive techno-economic comparison, environmental impact assessment, and a critical discussion of the findings within the context of residential solar deployment in the Kurdistan Region of Iraq.

4.2 Analysis of Measured PV Performance Data

4.2.1 Diurnal Power Output Patterns

Analysis of the 21 field measurements collected across 7 days (Table 3.6) reveals distinct diurnal and seasonal performance patterns. Table 4.1 summarizes the average daily power output by time of day.

Table 4.1: Average Measured Power Output (P_{max}) by Time of Day

Period	**Morning (Wp)**	**Midday (Wp)**	**Afternoon (Wp)**
Winter (12–16 Feb)	420.1	452.6	534.2
Spring (13 Apr)	522.6	727.4	378.7*
Overall Average	**453.5**	**533.3**	**478.3**

*Spring afternoon average influenced by the anomalous 6°C/155 Wp data point (see Section 3.6.4).

Key observations from the measured data:

1. The highest single power output recorded was **832.4 Wp** (13 April, midday, Session 1), corresponding to favorable spring irradiance conditions at 15°C ambient temperature.
2. The lowest valid output was **35.3 Wp** (12 February, morning), reflecting early-morning low-altitude solar conditions during winter.
3. The spring measurements demonstrate higher peak outputs than winter, consistent with increased solar altitude and longer day length.
4. Average midday output exceeds morning and afternoon output in most cases, confirming the expected diurnal irradiance curve.

4.2.2 Temperature Effects on Performance

The field data confirms the negative correlation between ambient temperature and PV output voltage. Table 4.2 summarizes the observed relationship.

Table 4.2: Observed Temperature–Voltage Relationship

Temp. Range (°C)	**Avg. V_{oc} (V)**	**Avg. V_{mp} (V)**
10–12	46.8	40.8
13–15	46.7	40.7
16–19	45.9	39.3

While the temperature range during the measurement campaign (10–19°C) was relatively narrow, a discernible voltage reduction of approximately 0.9 V in V_{oc} and 1.5 V in V_{mp} was observed between the coolest and warmest conditions. Extrapolating using the module's temperature coefficient of –0.34%/°C, summer conditions (45–50°C) would result in estimated power losses of **7–9%** relative to STC ratings, consistent with literature values of 15–20% for extreme Middle Eastern summer conditions [44].

4.2.3 Wind Speed Effects

Wind speed during the measurement campaign ranged from 2 to 25 km/h. Moderate wind speeds (10–18 km/h) correlated with marginally higher power outputs due to convective cooling of module surfaces. The highest recorded output day (16 February, midday: 702 Wp) coincided with a wind speed of 25 km/h, supporting the beneficial cooling effect documented in the literature.

4.2.4 Fill Factor Analysis

The Fill Factor (FF) is a key indicator of cell quality and was calculated for each measurement point using:

$$FF = \frac{P_{max}}{V_{oc} \times I_{sc}} = \frac{V_{mp} \times I_{mp}}{V_{oc} \times I_{sc}} \tag{4.1}$$

Table 4.3: Calculated Fill Factor Values for Selected Measurements

Date	**Time**	$V_{oc} \times I_{sc}$	P_{max}	**FF**	**Assessment**
13-2 Afternoon	15:00	709.5	709.6	1.00†	Near-unity
15-2 Morning	08:00	777.4	650.0	0.84	Excellent
16-2 Midday	12:00	829.5	702.0	0.85	Excellent
13-4 S1 Midday	12:00	1037.3	832.4	0.80	Good
13-4 S2 Midday	12:00	711.5	622.4	0.88	Excellent
Average FF (valid points)				**0.82**	Good–Excellent

†FF $\geq$ 1.0 indicates measurement uncertainty; see Section 3.6.4 limitations.

The average Fill Factor of **0.82** is consistent with high-quality monocrystalline silicon modules (typical range: 0.75–0.85) and confirms that the installed modules are performing within manufacturer specifications [11].

4.3 Energy Yield Estimation

4.3.1 Daily Energy Production

Using the measured power data, the estimated daily energy yield was calculated by integrating the average power output over the effective sunshine hours. For Erbil, the effective sunshine

duration averages 5.0 Peak Sun Hours (PSH) annually:

$$E_{day} = P_{avg} \times PSH_{effective} \tag{4.2}$$

Based on the overall average measured power output of 488.4 Wp from the field campaign:

$$E_{day} = 488.4 \times 5.0 = 2{,}442 \text{ Wh/day} \approx 2.44 \text{ kWh/day (per module equivalent)} \tag{4.3}$$

For the full 9-module array (4.95 kWp system), scaling proportionally and accounting for system losses:

Table 4.4: Estimated Daily and Annual Energy Production

Parameter	**On-Grid**	**Hybrid**
Daily Energy (Winter)	16.8 kWh	15.9 kWh
Daily Energy (Summer)	22.5 kWh	21.3 kWh
Daily Energy (Annual Avg.)	19.6 kWh	18.6 kWh
Annual Energy Production	7,154 kWh	6,789 kWh
Energy Delivered to Load	7,154 kWh	6,789 kWh
Inverter Efficiency	97.5%	96.5%
Battery Round-Trip Losses	–	5–8%

The hybrid system produces approximately **5.1% less** delivered energy annually due to battery charging/discharging losses. However, this deficit is offset by the significantly higher utilization of generated energy, as discussed in Section 4.4.4.

4.4 Performance Metrics Comparison

4.4.1 Specific Yield

The specific yield measures energy generated per unit of installed capacity:

$$Y_{sp} = \frac{E_{annual}}{P_{rated}} \tag{4.4}$$

- **On-grid:** $Y_{sp} = \frac{7{,}154}{4.95} = 1{,}445$ kWh/kWp/year

- **Hybrid:** $Y_{sp} = \frac{6{,}789}{4.95} = 1{,}372$ kWh/kWp/year

Both values fall within the expected range for the Kurdistan Region (1,300–1,600 kWh/kWp/year) and compare favorably with regional benchmarks [9].

4.4.2 Performance Ratio

The Performance Ratio indicates overall system quality:

$$PR = \frac{E_{actual}}{P_{rated} \times PSH \times 365} \tag{4.5}$$

- **On-grid:** $PR = \frac{7{,}154}{4.95 \times 5.0 \times 365} = \frac{7{,}154}{9{,}034} = 0.792$ (**79.2%**)
- **Hybrid:** $PR = \frac{6{,}789}{9{,}034} = 0.751$ (**75.1%**)

The on-grid PR of 79.2% is within the typical range of 72–85% reported for grid-connected PV systems in Iraq's hot arid climate. The hybrid system's lower PR reflects the additional battery conversion losses.

4.4.3 Capacity Factor

$$CF = \frac{E_{annual}}{P_{rated} \times 8{,}760} \times 100\% \tag{4.6}$$

- **On-grid:** $CF = \frac{7{,}154}{4.95 \times 8{,}760} = \frac{7{,}154}{43{,}362} = 16.5\%$
- **Hybrid:** $CF = \frac{6{,}789}{43{,}362} = 15.7\%$

These capacity factors fall within the 14–18% range reported for PV systems in Iraq, confirming that the system performs consistently with regional expectations.

4.4.4 Self-Consumption and Self-Sufficiency

Table 4.5: Self-Consumption and Self-Sufficiency Comparison

Metric	On-Grid	Hybrid
Self-Consumption Ratio (SCR)	30–40%	70–80%
Self-Sufficiency Ratio (SSR)	35–45%	55–65%
Excess Energy Exported	60–70%	20–30%
Grid Dependency	High	Medium–Low
Backup During Outages	None	6–10 hours

The hybrid system achieves approximately **double** the self-consumption ratio of the on-grid system. This is a decisive advantage in the Kurdistan Region where net metering policies are nascent and export compensation is negligible. The hybrid system's ability to store and self-consume 70–80% of generated energy, compared to only 30–40% for on-grid, translates directly into greater economic value per kWh generated [21].

4.5 Techno-Economic Analysis

4.5.1 Capital Cost Comparison

Table 4.6: Itemized Capital Cost Comparison (USD, 2026 Estimates)

Component	On-Grid ($)	Hybrid ($)
PV Modules (9 × 550Wp)	1,350	1,350
Inverter (5 kW)	600	1,200
Battery (48V 200Ah LFP)	–	1,800
Mounting Structure	400	400
Wiring & Protection	250	350
Installation Labor	300	450
Bidirectional Meter	100	100
Total CAPEX	**$3,000**	**$5,650**
Cost per Watt	**$0.61/Wp**	**$1.14/Wp**

The hybrid system commands an **88% cost premium** over the on-grid configuration, driven primarily by the battery bank ($1,800) and the more sophisticated hybrid inverter ($1,200 vs. $600). These costs reflect 2026 local market prices in Erbil following Iraq's reduction of solar equipment import duties from 33% to 5% in March 2026.

4.5.2 Annual Savings Calculation

Annual savings are calculated based on two revenue streams: avoided grid electricity costs and avoided diesel generator subscription costs.

Table 4.7: Annual Savings Breakdown

Savings Component	**On-Grid ($/yr)**	**Hybrid ($/yr)**
Grid electricity offset	285	428
Diesel generator avoided	0	1,200
Exported energy credit	45	15
Total Annual Savings	**$330**	**$1,643**

Grid tariff: $0.015/kWh (subsidized); Diesel generator: $100–150/month (10–15A subscription) [40].

The hybrid system generates **5× greater annual savings** than the on-grid system, primarily because it displaces expensive diesel generator subscriptions during grid outages. The on-grid system cannot capture this value because it shuts down during outages.

4.5.3 Payback Period

$$SPP = \frac{C_{total}}{S_{annual}} \tag{4.7}$$

- **On-grid:** $SPP = \frac{3{,}000}{330} = 9.1$ years
- **Hybrid:** $SPP = \frac{5{,}650}{1{,}643} = 3.4$ years

Despite its higher capital cost, the hybrid system achieves a payback period of only **3.4 years**—less than half the on-grid system's 9.1 years. This is driven by the substantial diesel generator displacement savings. These results are consistent with the 1–7 year payback range reported for Iraqi residential solar installations.

4.5.4 Levelized Cost of Electricity (LCOE)

Using a discount rate of 8%, system lifetime of 25 years, annual O&M cost of 1% of CAPEX, and 0.5%/year degradation:

$$LCOE = \frac{C_{total} + \sum_{t=1}^{25} \frac{C_{O\&M}}{(1+0.08)^t}}{\sum_{t=1}^{25} \frac{E_t}{(1+0.08)^t}} \tag{4.8}$$

Table 4.8: LCOE Comparison with Alternative Energy Sources

Energy Source	**LCOE ($/kWh)**
On-Grid Solar PV (this study)	0.052
Hybrid Solar PV (this study)	0.098
Iraqi Grid (subsidized tariff)	0.015
Diesel Generator (private)	0.32–0.45
Global Utility Solar Benchmark	0.039

Both PV configurations produce electricity at a fraction of the cost of diesel generators. The on-grid LCOE of **$0.052/kWh** and hybrid LCOE of **$0.098/kWh** are both significantly below the diesel generator cost of $0.32–0.45/kWh, confirming the strong economic case for residential solar in Erbil [2].

4.5.5 Battery Replacement Economics

The $LiFePO_4$ battery (3,000–10,000 cycle life) is expected to require one replacement during the 25-year system lifetime, at approximately Year 12–15. Assuming a replacement cost of $1,200 (accounting for projected battery price declines of 5–8% per year):

- Discounted replacement cost at Year 13: $\frac{1{,}200}{(1.08)^{13}} = \$441$
- Adjusted hybrid LCOE (with replacement): approximately **$0.106/kWh**

Even with battery replacement, the hybrid system remains **3× cheaper** than diesel generation.

4.6 Environmental Impact Assessment

4.6.1 CO_2 Emission Reduction

The CO_2 emissions avoided by each system configuration were calculated using standard emission factors:

- Iraqi grid emission factor: 0.85 kg CO_2/kWh [46]
- Diesel generator emission factor: 0.70 kg CO_2/kWh

Table 4.9: Annual CO_2 Emission Reductions

Parameter	**On-Grid**	**Hybrid**
Grid electricity displaced (kWh)	2,862	4,281
Diesel generation displaced (kWh)	0	3,500
CO_2 avoided (grid) (kg/yr)	2,433	3,639
CO_2 avoided (diesel) (kg/yr)	0	2,450
Total CO_2 avoided (kg/yr)	**2,433**	**6,089**
25-year lifetime (tonnes)	**60.8**	**152.2**

The hybrid system avoids **2.5× more CO_2 emissions** than the on-grid system over its lifetime, primarily through diesel generator displacement. Over 25 years, a single hybrid residential installation prevents an estimated **152 tonnes of CO_2** from entering the atmosphere.

4.6.2 Air Quality Benefits

Beyond CO_2, displacing diesel generators eliminates emissions of:

- Nitrogen oxides (NO_x): 5–10 g/kWh
- Particulate matter ($PM_{2.5}$): 0.3–0.6 g/kWh
- Sulfur dioxide (SO_2): 1–3 g/kWh
- Carbon monoxide (CO): 2–5 g/kWh
- Noise pollution: 75–95 dB at source

For a household displacing 3,500 kWh/year of diesel generation, this represents an annual reduction of approximately 17–35 kg of NO_x, 1–2 kg of $PM_{2.5}$, and complete elimination of generator noise during nighttime hours—a significant improvement in residential quality of life.

4.7 Comparative Summary

Table 4.10: Comprehensive On-Grid vs. Hybrid Comparison

Parameter	**On-Grid**	**Hybrid**	**Advantage**
Capital Cost	$3,000	$5,650	On-Grid
Annual Energy Yield	7,154 kWh	6,789 kWh	On-Grid
Performance Ratio	79.2%	75.1%	On-Grid
Self-Consumption Ratio	30–40%	70–80%	Hybrid
Self-Sufficiency Ratio	35–45%	55–65%	Hybrid
Annual Savings	$330	$1,643	Hybrid
Payback Period	9.1 years	3.4 years	Hybrid
LCOE	$0.052/kWh	$0.098/kWh	On-Grid
Outage Backup	None	6–10 hours	Hybrid
CO_2 Avoided (25 yr)	60.8 tonnes	152.2 tonnes	Hybrid
Diesel Displacement	None	Full	Hybrid
Overall Suitability	**Limited**	**Highly Suitable**	**Hybrid**

4.8 Discussion

4.8.1 Interpretation of Key Findings

The results of this study reveal a critical insight: *in contexts where grid reliability is low and diesel generator dependency is high, the conventional economic advantage of on-grid PV systems is inverted.* While on-grid systems demonstrate superior technical performance metrics (higher PR, CF, and energy yield) and lower LCOE in absolute terms, these advantages are rendered largely academic in the Kurdistan Region context for three reasons:

1. **On-grid systems cannot operate during grid outages,** which occur daily for 6–12 hours in Erbil. During these critical hours, households must either endure power cuts or subscribe to diesel generators—the very problem solar PV is intended to solve.

2. **The economic value of energy is not uniform.** A kWh consumed during a grid outage (displacing diesel at \$0.32–0.45/kWh) is worth 20–30× more than a kWh displacing subsidized grid electricity at \$0.015/kWh. The hybrid system captures this high-value displacement; the on-grid system cannot.

3. **Without mature net metering,** the 60–70% of on-grid generation that is exported to the grid generates minimal or zero revenue, dramatically reducing the system's effective economic return.

4.8.2 Comparison with Published Literature

The findings align with and extend several key studies:

- The hybrid system's self-consumption ratio of 70–80% is consistent with Koskela et al. [21], who reported that battery storage increases self-consumption by over 30 percentage points.
- The payback period of 3.4 years for the hybrid system is within the 1–7 year range reported for Iraqi residential solar installations, and compares favorably with Shah et al. [43] and Gabr et al. [8].
- The Performance Ratio of 79.2% aligns with the 72–85% range documented for hot arid climates [44].
- The Fill Factor average of 0.82 confirms module quality consistent with Green et al. [11].

4.8.3 Sensitivity to Key Variables

The economic analysis is most sensitive to:

- **Diesel generator subscription cost:** If generator costs decrease by 50%, hybrid payback extends to 5.2 years—still competitive.
- **Grid reliability improvement:** If grid availability reaches 22+ hours/day, the diesel displacement value diminishes, making on-grid more competitive.
- **Battery cost trajectory:** Continued LFP battery price reductions (projected 5–8%/year) will further improve hybrid economics.
- **Net metering implementation:** Effective net metering would significantly improve on-grid economics by monetizing exported energy.

4.8.4 The "Runaki" (Light) Project: Impact on Residential Solar Economics

The "Runaki" project, launched by the Kurdistan Regional Government's Ministry of Electricity in late 2024, represents the most significant structural reform of the Kurdistan Region's electricity sector in decades. The project aims to deliver 24-hour continuous national grid electricity to all households across the Kurdistan Region by the end of 2026, progressively replacing dependence on private neighborhood diesel generators [22]. While the project carries substantial potential benefits, it has also generated considerable public controversy—particularly regarding its progressive tiered tariff structure—with direct implications for residential solar PV economics.

Progressive Tiered Tariff Structure

Under the Runaki framework, residential electricity is billed according to a progressive (tiered) pricing model, where the per-kWh rate increases significantly as monthly consumption rises:

Table 4.11: Runaki Project: Progressive Residential Electricity Tariff Structure

Consumption Tier	Rate (IQD/kWh)	Rate ($/kWh)	Multiplier
1 – 400 kWh	72	0.055	1.0× (base)
401 – 800 kWh	108	0.082	1.5×
801 – 1,200 kWh	175	0.134	2.4×
1,201 – 1,600 kWh	265	0.203	3.7×
1,601+ kWh	350	0.268	4.9×

Exchange rate: 1 USD ≈ 1,310 IQD (2026). Source: KRG Ministry of Electricity.

This structure means that a household consuming 2,000 kWh/month—typical during Erbil's extreme summer when air conditioning operates 8–12 hours daily—would face a blended cost approximately **2.5–3× higher per kWh** than a household consuming only 400 kWh/month. The top-tier rate of 350 IQD/kWh ($0.268/kWh) approaches the cost of diesel generator electricity ($0.32–0.45/kWh), undermining the intended cost advantage of the national grid for high-consumption households.

Benefits of the Runaki Project

1. **24-hour grid reliability:** The most significant benefit is the promise of continuous electricity supply, eliminating the chronic 6–12 hour daily outages that have plagued the Kur-

distan Region for decades.

2. **Diesel generator displacement:** As grid availability approaches 24 hours, households can reduce or eliminate their diesel generator subscriptions (100,000–450,000 IQD/month), yielding substantial savings for moderate-consumption households.

3. **Environmental improvement:** Eliminating thousands of neighborhood diesel generators reduces CO_2, NO_x, particulate matter, and noise pollution across residential areas.

4. **Infrastructure modernization:** The project includes smart metering infrastructure, enabling accurate consumption tracking and laying the groundwork for future net metering and solar prosumer policies.

5. **Conservation incentive:** The tiered structure encourages energy efficiency and discourages wasteful consumption patterns.

Disadvantages and Public Concerns

1. **Escalating bills for high-consumption households:** The most widespread complaint is that households with high electricity demand—particularly during summer—face bills that are **2–3× higher** than anticipated. A household consuming 1,500 kWh/month would pay approximately 194,500 IQD ($148), while a household consuming 2,000 kWh/month would pay approximately 369,500 IQD ($282). Many citizens report that these bills exceed their previous combined grid and generator costs.

2. **Disproportionate impact on low-income families:** Critics, including several Kurdistan Region parliamentarians, have argued that the tiered structure places an unfair burden on low- and middle-income households, particularly large families with higher baseline consumption needs.

3. **Meter reading irregularities:** Public complaints have emerged regarding billing periods extending beyond the standard 30 days (e.g., 45-day billing cycles), which artificially inflates consumption into higher tariff tiers.

4. **Seasonal penalty:** The tiered structure disproportionately penalizes summer consumption, when cooling is not a luxury but a health necessity in temperatures exceeding 43–50°C.

5. **Absence of prosumer framework:** The Runaki project does not currently include provisions for net metering or solar prosumer compensation, meaning households with PV systems cannot sell excess generation back to the grid at favorable rates.

6. **Legal challenges:** Political opposition factions attempted to halt the project through the Iraqi Federal Supreme Court in 2025, though the court dismissed the case citing lack of jurisdiction over regional administrative policy.

Implications for Residential Solar PV Systems

The Runaki tariff structure fundamentally alters the economic calculus for residential solar PV in the Kurdistan Region:

1. **Solar becomes more valuable under tiered pricing:** Under the old flat-rate subsidized tariff ($0.015/kWh), solar PV displaced very cheap electricity, resulting in long payback periods. Under Runaki's tiered structure, every kWh of solar self-consumption that keeps a household below the 400 kWh threshold avoids electricity priced at 2.4–4.9× the base rate. This dramatically improves the economic return of solar PV.

2. **Revised payback calculation under Runaki tariffs:** Using the Runaki tiered rates instead of the flat subsidized rate, the annual grid electricity savings increase substantially:

Table 4.12: Revised Annual Savings Under Runaki Tiered Tariffs

Component	**On-Grid ($/yr)**	**Hybrid ($/yr)**
Grid savings (Runaki tariff)	580	870
Diesel generator avoided	0	600†
Total Annual Savings	**$580**	**$1,470**
Revised Payback	**5.2 years**	**3.8 years**

†Reduced diesel displacement value assumes Runaki achieves 20–22 hours daily grid availability.

Under Runaki tariffs, the on-grid payback improves from 9.1 to **5.2 years**, while the hybrid payback remains competitive at **3.8 years** even with reduced diesel displacement value.

3. **Hybrid systems retain their advantage:** Even if Runaki achieves its goal of 24-hour supply, the tiered pricing structure ensures that hybrid systems—which maximize self-consumption and minimize grid imports into higher tiers—remain economically superior for high-consumption households.

4. **Solar as a "tariff shield":** Residential solar PV effectively acts as a hedge against the escalating tiered tariff structure, keeping households in lower billing tiers and providing protection against potential future tariff increases.

4.8.5 Limitations of the Analysis

1. The field measurements were conducted over 7 days in two seasons (winter and spring); summer performance was projected rather than directly measured, introducing estimation uncertainty for the highest-demand period.

2. The data anomalies identified in Section 3.6.4 (6°C April temperature, inverted Feb 14 output, $I_{mp} = I_{sc}$ on Feb 12) may introduce minor distortions in aggregate calculations.

3. Economic projections assume stable grid tariffs and diesel prices; actual future costs may vary due to subsidy reforms or market fluctuations.

4. Battery degradation modeling uses manufacturer specifications; actual field degradation in Erbil's extreme summer temperatures may differ.

4.9 Chapter Summary

This chapter presented the complete results and analysis of the study:

1. Field-measured data confirmed that the 4.95 kWp PV system performs within expected parameters, with a Fill Factor of 0.82, specific yields of 1,372–1,445 kWh/kWp/year, and Performance Ratios of 75–79%.

2. The hybrid system achieves a self-consumption ratio of 70–80% versus 30–40% for on-grid, representing a decisive advantage in the absence of net metering.

3. The hybrid system achieves a payback period of **3.4 years** versus 9.1 years for on-grid, driven by diesel generator displacement savings of approximately $1,200/year.

4. Both systems produce electricity far below the cost of diesel generation ($0.052–0.098/kWh vs. $0.32–0.45/kWh).

5. The hybrid system avoids **152 tonnes of CO_2** over its 25-year lifetime, compared to 61 tonnes for on-grid.

6. For Erbil's context of unreliable grid supply and high diesel dependency, the **hybrid PV system is the clearly superior configuration** despite its higher capital cost.

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